MILITARY SPECIFICATION SHEET

TOOLS, FIBER OPTIC, STRIPPING, (50/125), (100/140), (400/430), (600/630), (1000/1040), TYPE X, COMPOSITION C

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the fiber optic tools described herein shall consist of this specification sheet and the latest issue of MIL-T-83523.

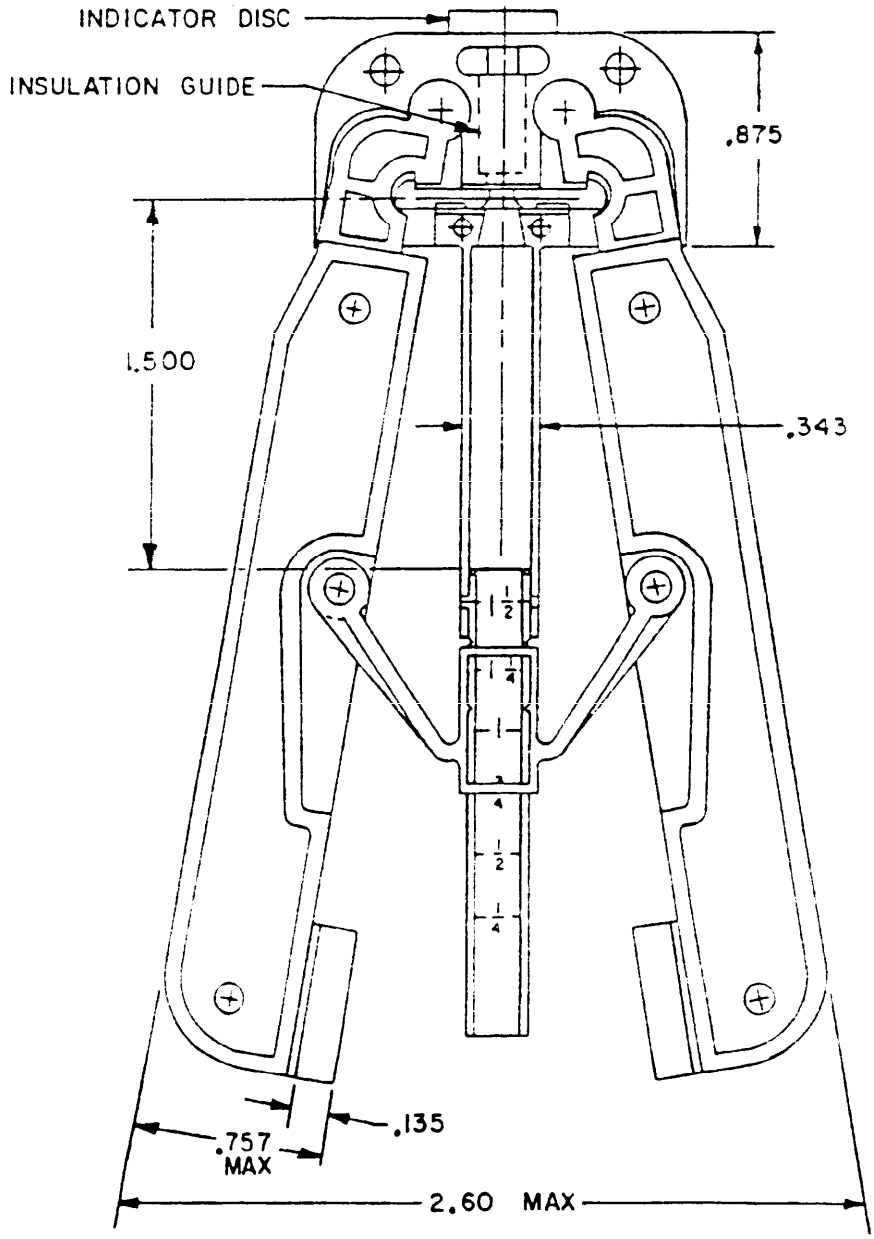
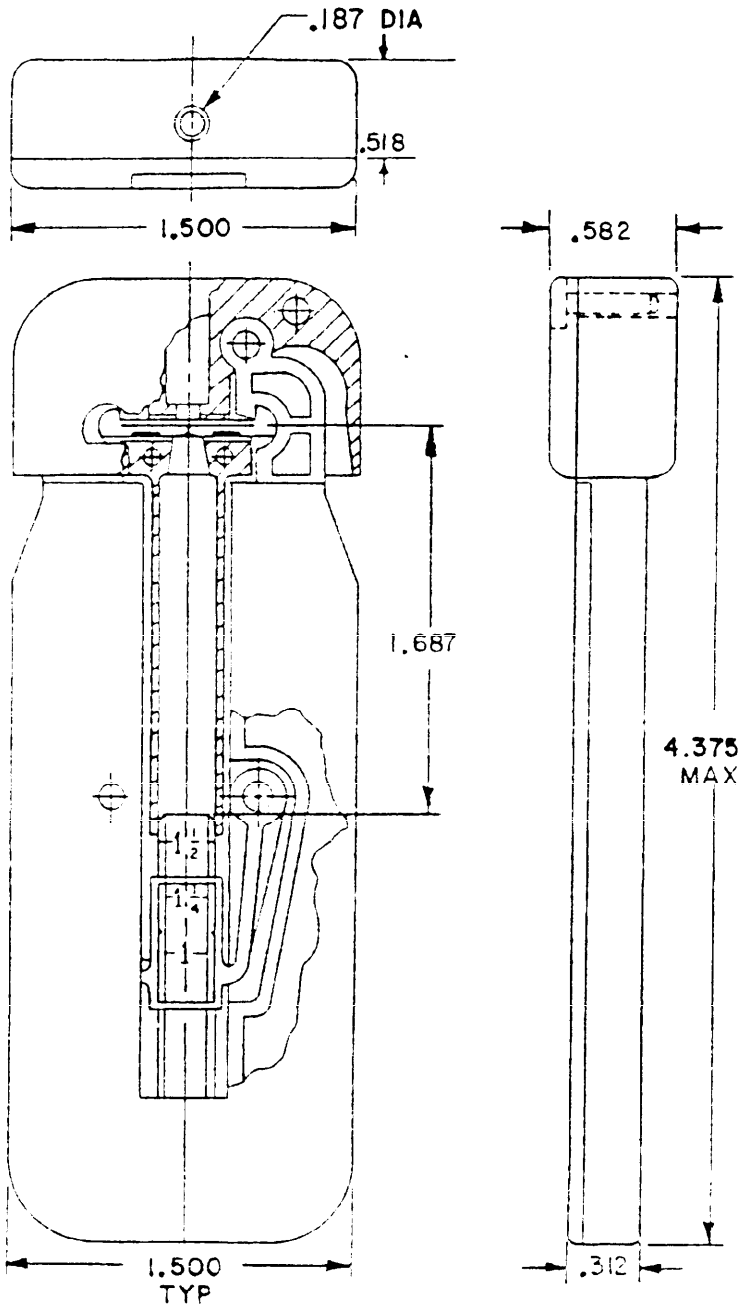


FIGURE 1a. Fiber stripping tool front view (open).

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Inches	mm
.135	3.43
.187	4.75
.312	7.92
.343	8.71
.518	13.16
.580	14.73
.582	14.78
.757	19.23
.875	22.22
1.500	38.10
1.687	42.85
2.437	61.90
2.60	66.04
4.375	111.12

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances shall be ±.010 (0.25 mm).

FIGURE 1b. Fiber stripping tool front view (closed).

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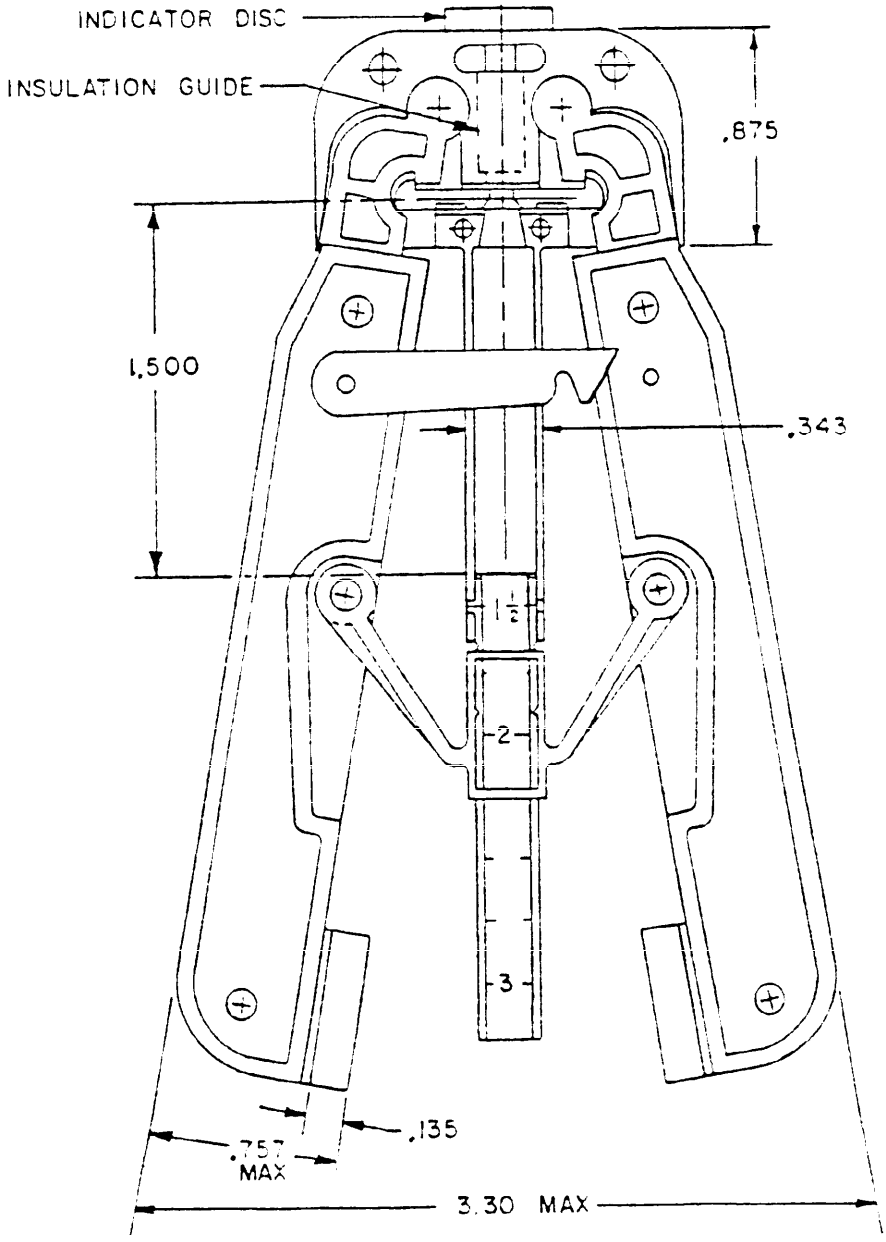
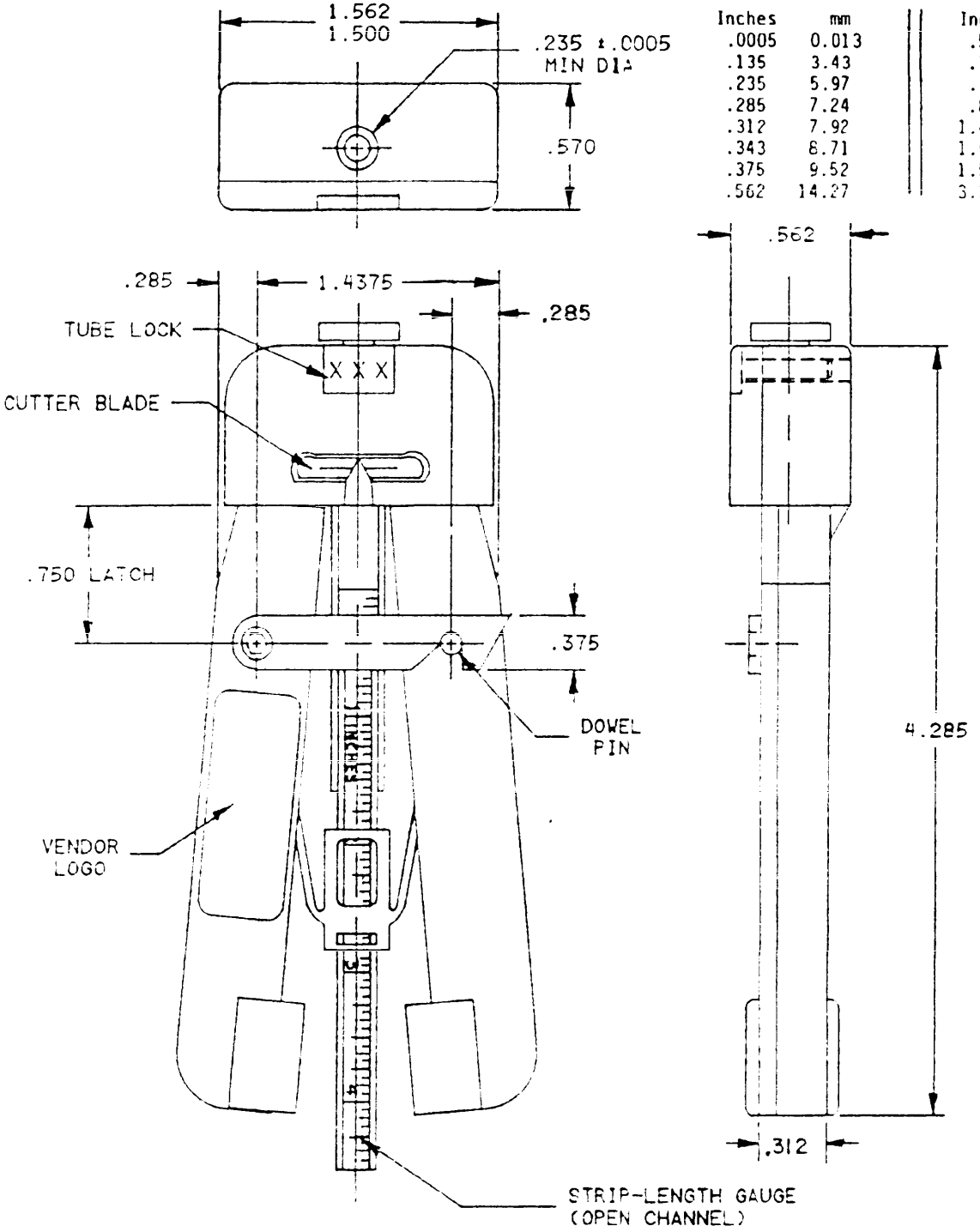


FIGURE 2a. Jacket stripping tool front view (open).

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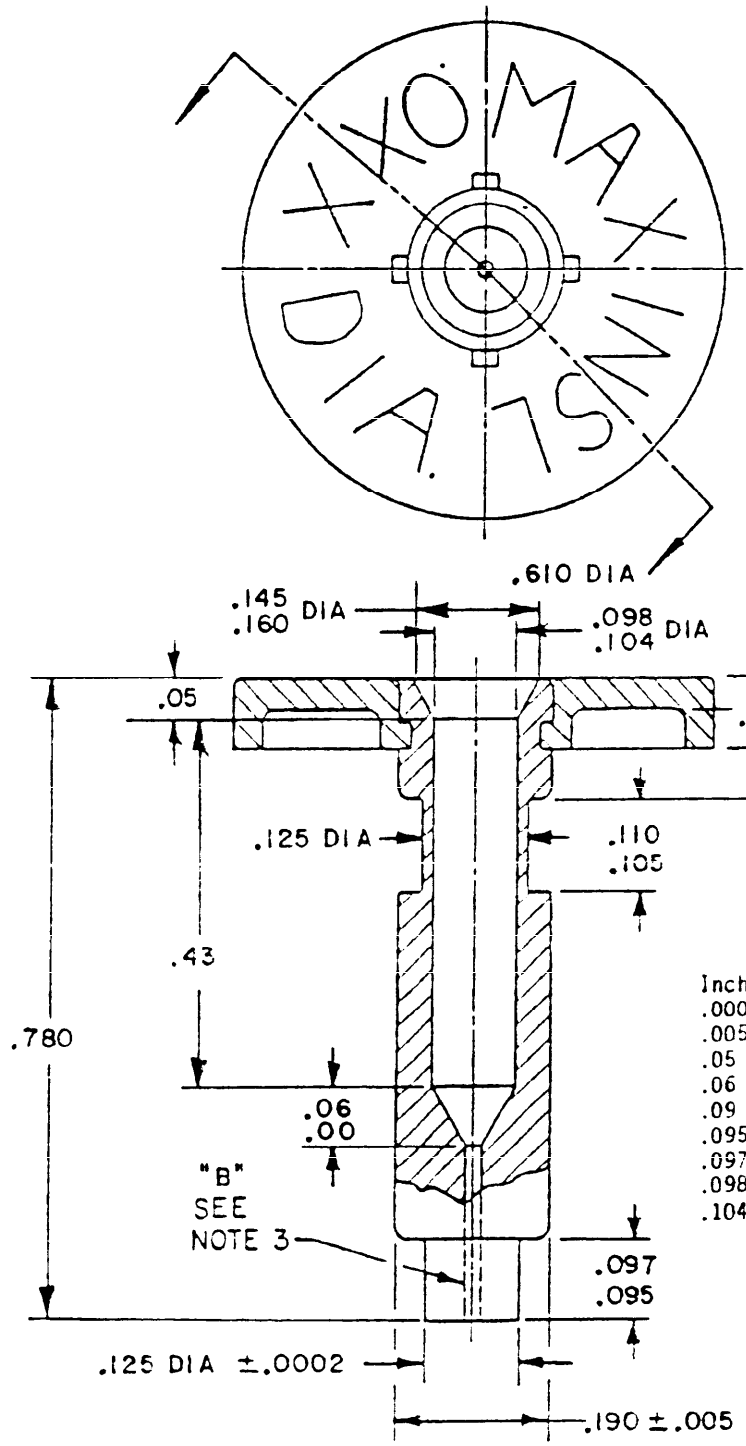


Inches	mm	Inches	mm
.0005	0.013	.570	14.48
.135	3.43	.750	19.05
.235	5.97	.757	19.23
.285	7.24	.875	22.22
.312	7.92	1.4375	36.512
.343	8.71	1.500	38.10
.375	9.52	1.562	39.67
.562	14.27	3.30	83.8

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances shall be ±.010 (0.25 mm).

FIGURE 2b. Jacket stripping tool (closed).

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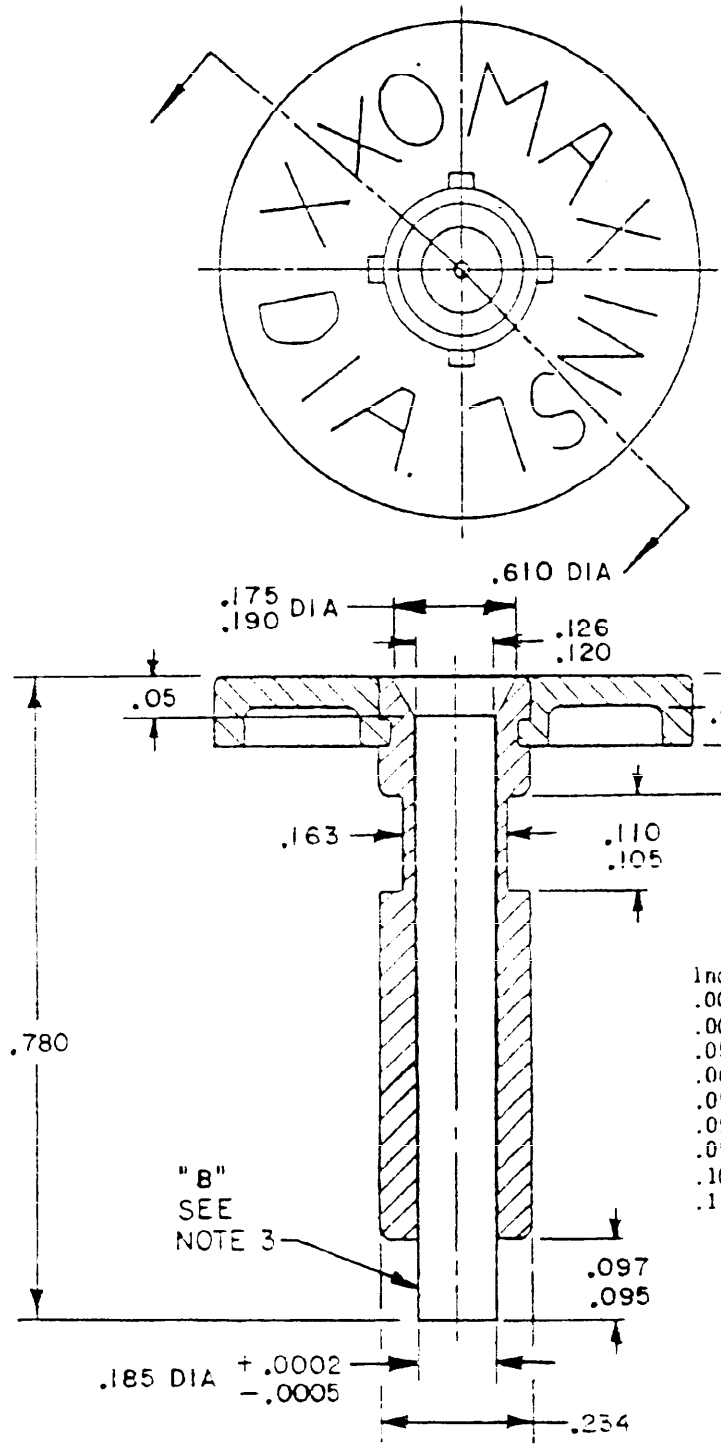


Inches	mm	Inches	mm
.0002	0.005	.105	2.67
.005	0.13	.110	2.79
.05	1.3	.125	3.18
.06	1.5	.145	3.68
.09	2.3	.160	4.06
.095	2.41	.190	4.83
.097	2.46	.43	10.92
.098	2.49	.610	15.49
.104	2.64	.780	19.81

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. See table I for diameter.
4. Unless otherwise specified, tolerances shall be ±.005 (0.13 mm) for three place decimals and ±.02 (0.51 mm) for two place decimals.

FIGURE 3. Fiber stripping tool insulation guide assembly.

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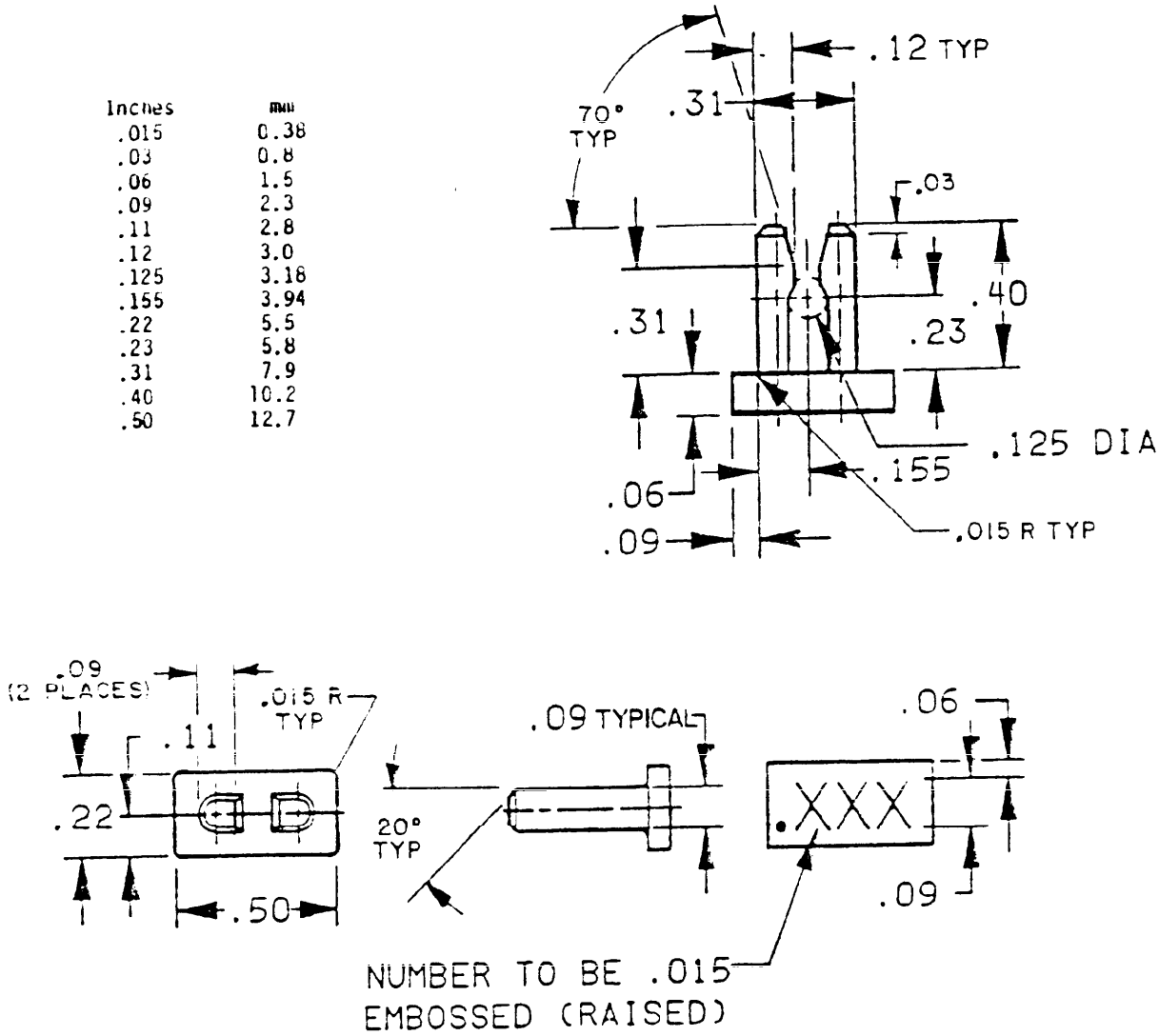
Inches	mm	Inches	mm
.0002	0.005	.120	3.05
.0005	0.013	.126	3.20
.05	1.3	.163	4.14
.06	1.5	.175	4.44
.090	2.29	.185	4.70
.095	2.41	.190	4.83
.097	2.46	.234	5.94
.105	2.67	.625	15.88
.110	2.79	.780	19.81

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. See table I for inside diameter.
4. Unless otherwise specified, tolerances shall be ±.005 (0.13 mm).

FIGURE 4. Jacket stripping tool insulation guide assembly.

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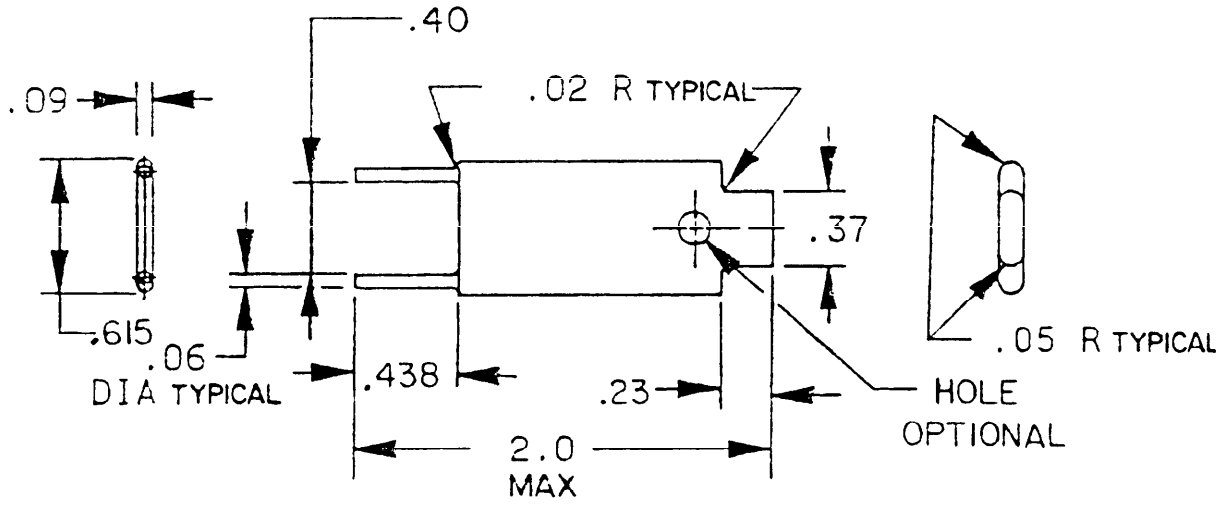
Inches	mm
.015	0.38
.03	0.8
.06	1.5
.09	2.3
.11	2.8
.12	3.0
.125	3.18
.155	3.94
.22	5.5
.23	5.8
.31	7.9
.40	10.2
.50	12.7



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances shall be ±.010 (0.25 mm) for three place decimals and ±.02 (0.5 mm) for two place decimals.

FIGURE 5. Guide tube lock.

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MATERIAL: PLASTIC

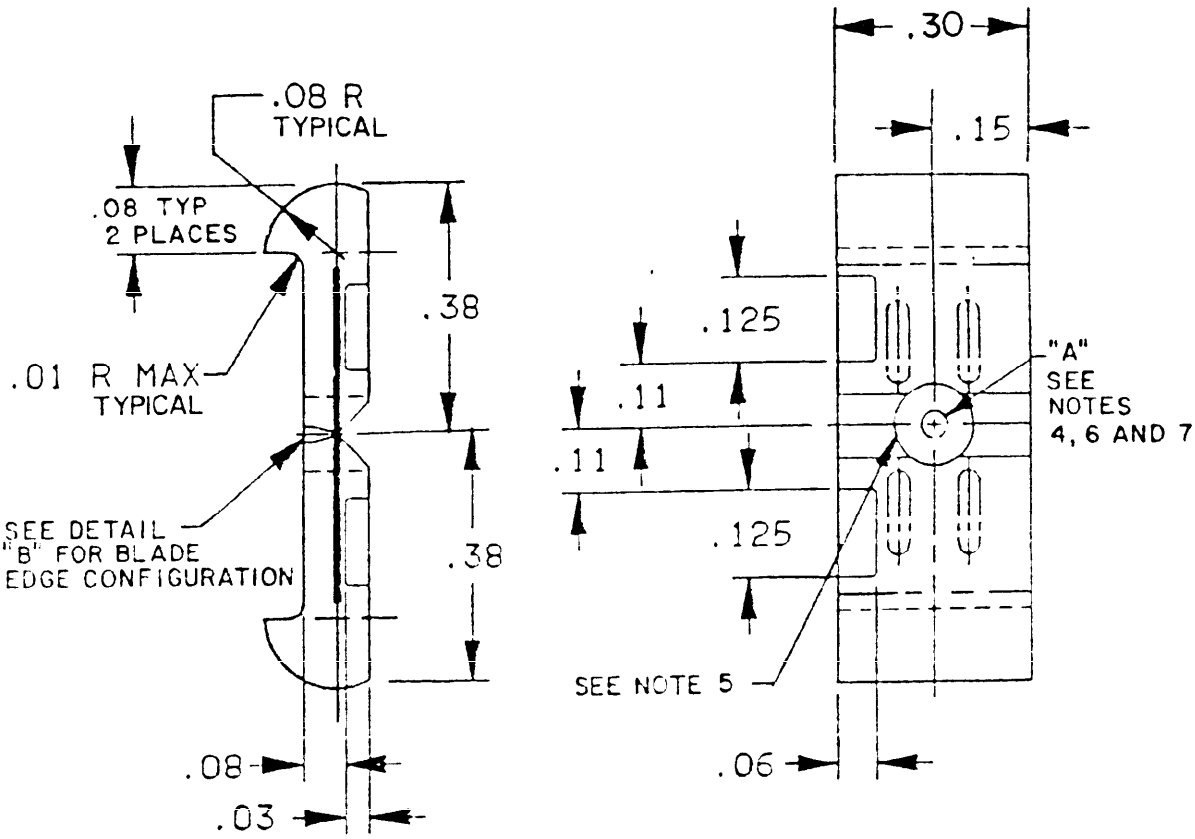
Inches	mm
.02	0.5
.05	1.3
.06	1.5
.09	2.3
.23	5.8
.37	9.4
.40	10.1
.438	11.12
.615	15.62
2.0	50.8

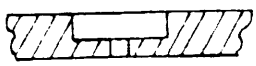
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances shall be ±.005 (0.13 mm) for three place decimals and ±.05 (1.3 mm) for two place decimals.

FIGURE 6. Blade push-out tool.

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Minimum recovery temperature is 250°F (121°C)





NORMAL:
Die-block blade:
Use for soft insulation.



SHARP:
Use for hard
insulations.

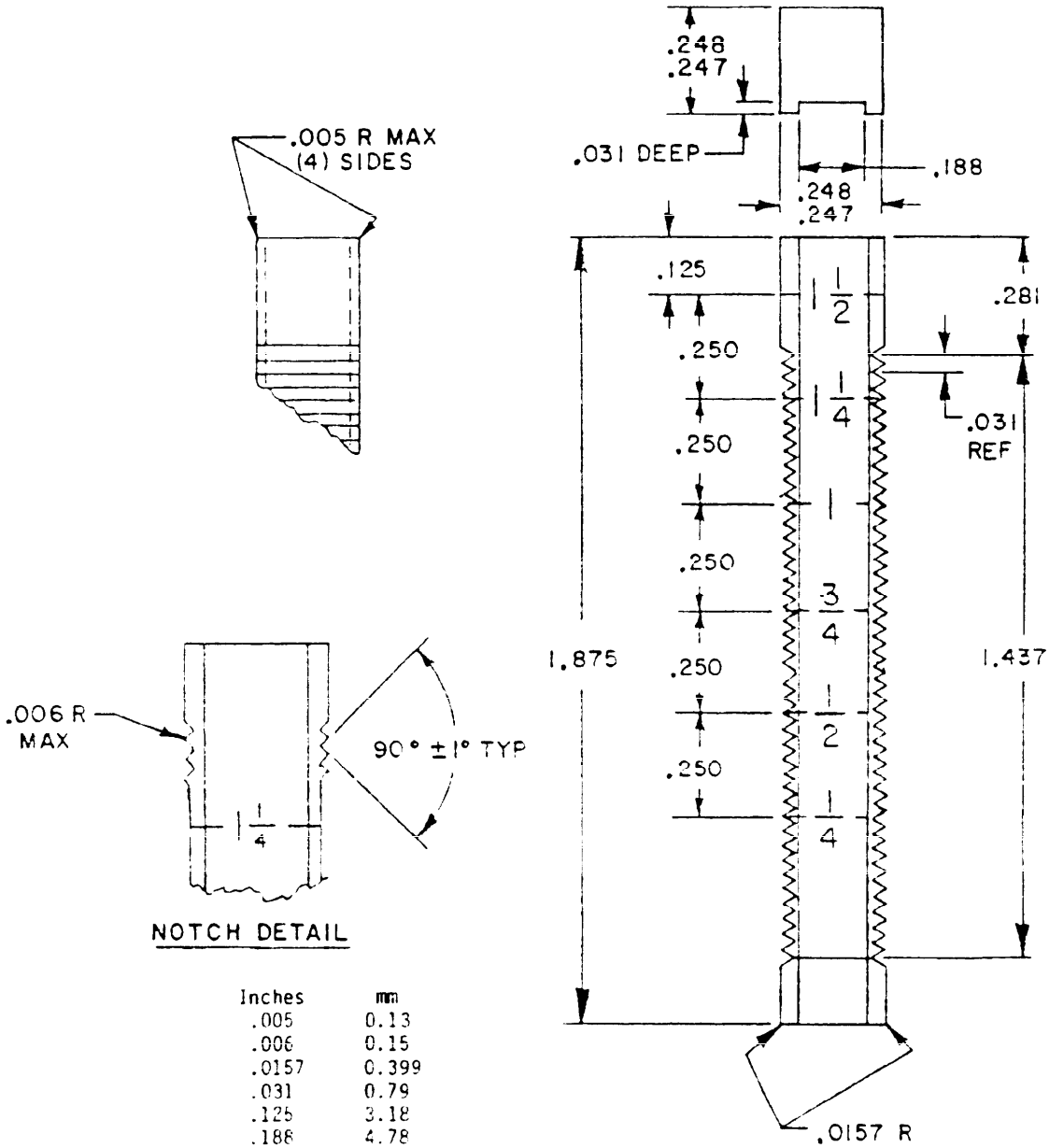
DETAIL "B"

Inches	mm
.01	0.3
.03	0.8
.06	1.5
.08	2.0
.11	2.8
.125	3.18
.15	3.8
.30	7.6
.38	9.7

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances shall be ±.005 (0.13 mm) for three place decimals and ±.02 (0.5 mm) for two place decimals.
4. Hole in blade and diameter must be concentric within .005 (0.13 mm).
5. Centerline noted must be parallel with "score mark" on blade within .0005 (0.012 mm).
6. .112 (2.84 mm) diameter of mold must be on center with hole diameter "A" of blade blank with .0005 (0.012 mm) concentricity on both sides for fiber stripping tool. For diameter of "A" see table I.
7. .185 (4.70 mm) diameter of mold must be on center with hole diameter "A" of blade blank with .0005 (0.012 mm) concentricity on both sides for jacket stripping tool. For diameter of "A" see table I.

FIGURE 7. Stripping blade pair.

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NOTCH DETAIL

Inches	mm
.005	0.13
.006	0.15
.0157	0.399
.031	0.79
.125	3.18
.188	4.78
.247	6.27
.248	6.30
.250	6.35
.281	7.14
1.437	36.50
1.875	47.62

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances shall be ±.005 (0.13 mm).

FIGURE 8. Strip length gauge.

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REQUIREMENTS:

Dimensions and configurations: See figures 1a, 1b, 2a, 2b, 3, 4, 5, 6, 7, 8, 9, 10, 11a, and 11b.

Weight: .5 pounds maximum.

Material:

Tool frame: MIL-M-14 or equivalent.

Blades: 1095 spring steel.

Insulation guide assembly: Aluminum and MIL-M-14 or equivalent.

Capability: Blades shall be replaceable. The tool shall be capable of stripping most sizes either by adjustment, blade replacement, or insulation guide substitution. The tool shall not nick the fiber during the stripping operation.

Part number: See table I.

Application: This tool shall be used to strip optical fibers as specified in DOD-C-85045/1 and DOD-F-49291.

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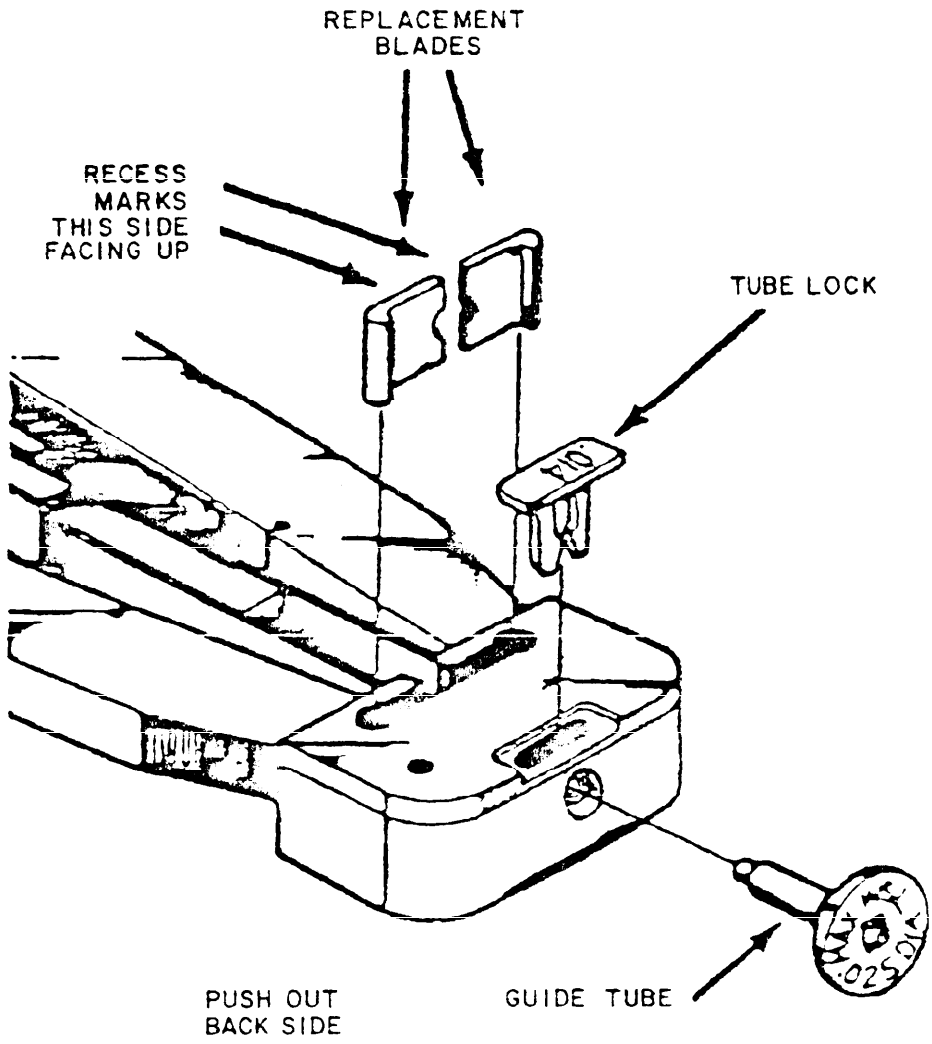
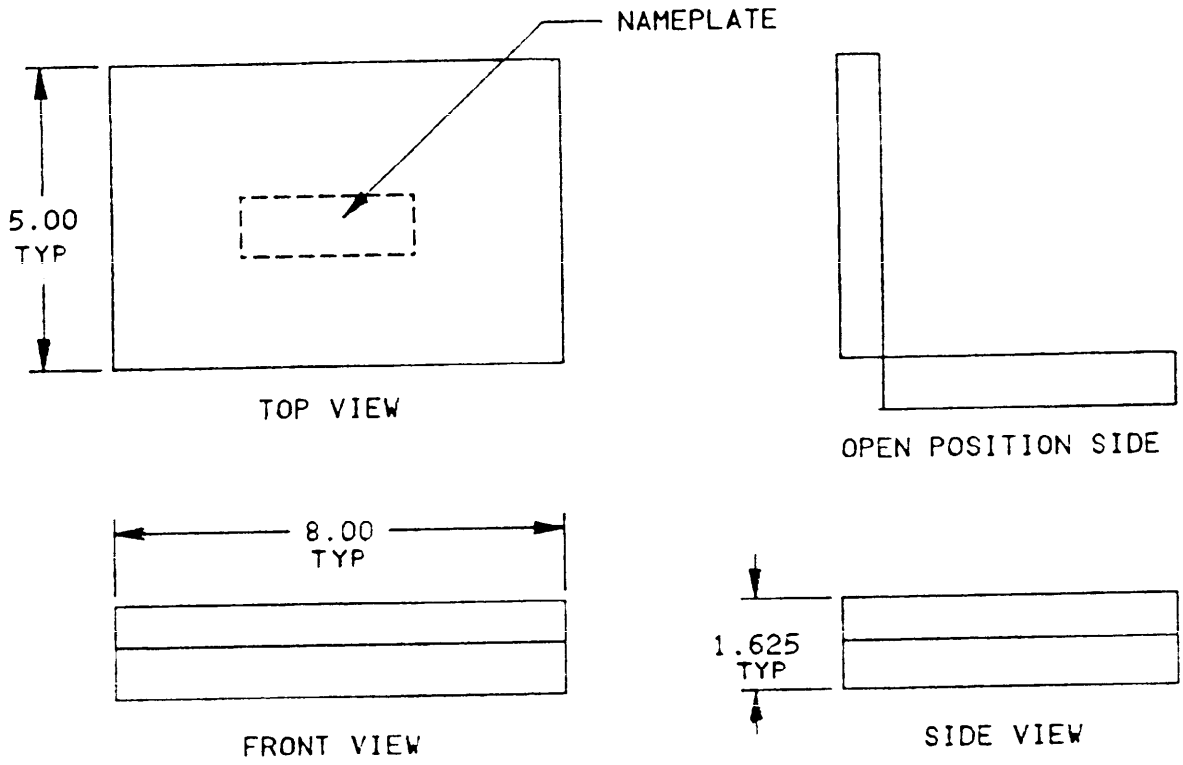


FIGURE 9. Fiber and jacket stripping tool assemblies.

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Inches	mm
1.625	41.28
5.00	127.0
8.00	203.2

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 10. Tool container.

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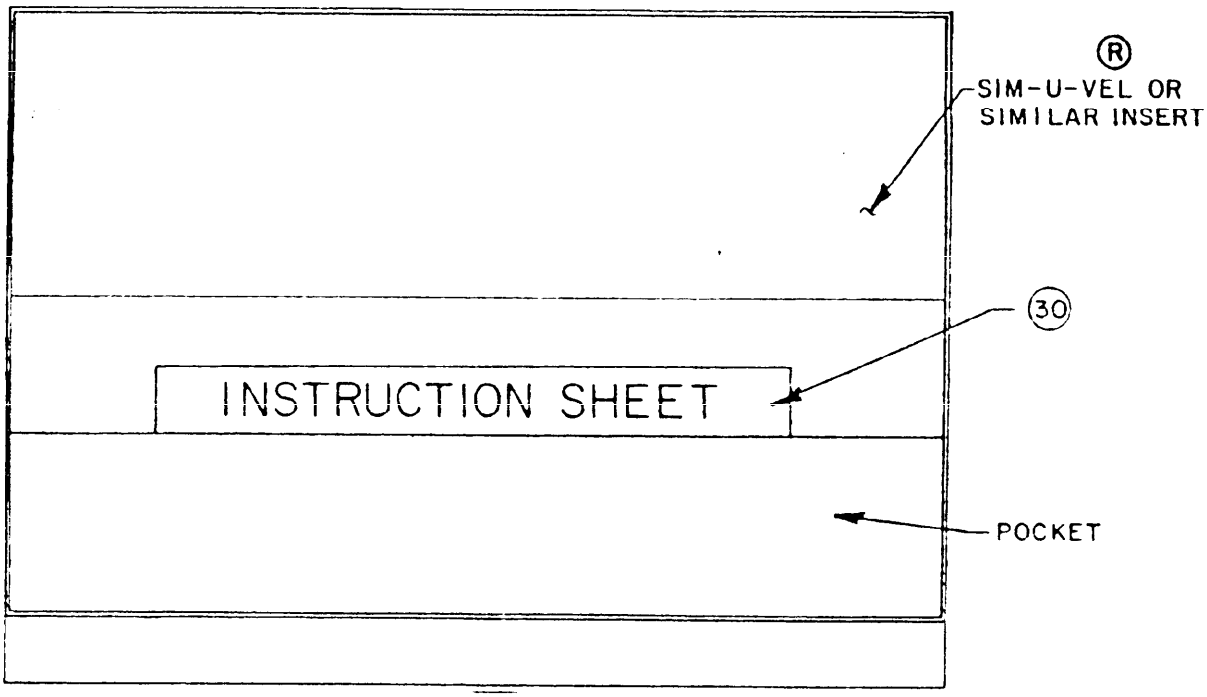


FIGURE 11a. Tool container cover.

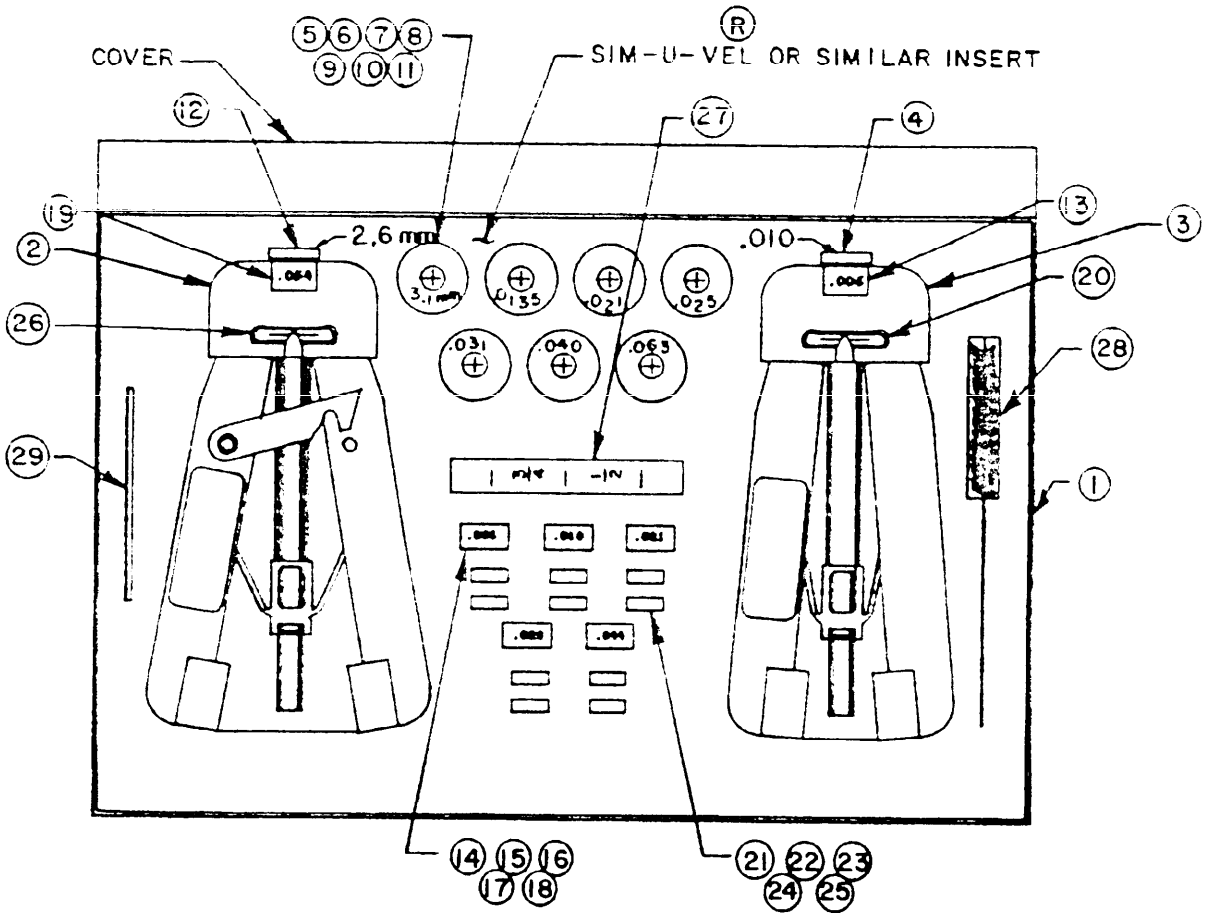


FIGURE 11b. Tool container base.

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TABLE I. Part list.

Qty reqd	MIL P/N M83523-16	Description	"B" diameter In (mm)	"A" Cutter diameter In (mm)	Fiber diameter µm	Color code	Item no.	Fig. no.
1	-04	Tool container				Blue	1	10
"	-05	Jacket stripping tool 1/				Light blue	2	2
"	-01	Fiber stripping tool 1/				Light blue	3	1
"	-46	Insulation guide assy.	.010 (.25)			White	4	3
"	-30	"	.0135 (.343)			"	6	3
"	-33	"	.021 (.53)			"	7	3
"	-34	"	.025 (.64)			"	8	3
"	-35	"	.031 (.79)			"	9	3
"	-37	"	.040 (1.02)			"	10	3
"	-41	"	.063 (1.60)			"	11	3
"	-47	"	.103 (2.62) 5/			White w/blue	12	4
"	-48	"	.122 (3.10) 5/			White w/blue	5	4
"	-50	Guide tube lock .006				Purple	13	5
"	-51	" .008				Red	14	5
"	-52	" .010				Light blue	15	5
"	-53	" .021				Royal blue	16	5
"	-54	" .028				Black	17	5
"	-56	" .044				Dark brown	18	5
"	-55	" .054				Natural	19	5

See footnotes at end of table.

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TABLE I. Part list - Continued.

Qty reqd	MIL P/N M83523-16	Description	"B" diameter In (mm)	"A" Cutter diameter In (mm)	Fiber diameter µm	Color code	Item no.	Fig. no.
1	-10	Stripping blade PR 2/ 3/		.006 (.15)	125 µm	Purple	20	7
"	-11	" 2/ 3/		.008 (.20)	125-150	Red	21	7
"	-12	" 2/ 3/		.010 (.25)	200-230	Light blue	22	7
"	-17	" 2/ 3/		.021 (.53)		Royal blue	23	7
"	-20	" 2/ 3/		.028 (.71)		Black	24	7
"	-24	" 2/ 3/		.044 (1.12) 4/	950-1000	Dark brown w/blue	25	7
"	-25	" 2/ 3/		.054 (1.37) 4/	1118-1140	Natural w/blue	26	7
"	-60	Insulation strip gauge				White	27	8
"	-02	Clean-out brush 6/				White	28	
"	-03	Push-out tool				White	29	6
"	-99	Complete tool set (includes all P/N's)						11
"		Instruction sheet 1/				White	30	

1/ Instruction sheet will be furnished with P/N M83523/16-01, P/N M83523/16-05 (individual stripping tools) as well as the complete set (M83523/16-99) when these items are ordered separately.

2/ Replacement blade package shall include matching tube lock.

3/ Cutter diameter tolerance ±.0002 (.005 mm).

4/ Sharp blade is indicated by a gold band and is only used with M83523/16-05, (see figure 7, detail "B").

5/ M83523/16-47 and M83523/16-48 to be used only with M83523/16-05.

6/ Approximate dimensions of clean out brush: Brush diameter .30"; brush length 1.25"; length of brush with handle 3.25". Brush material: Nylon or suitable substitute.

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P. Allocated and not used (see 3.10)

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Patent no.	Date issued	Term of patent (years)
3,827,317	August 6, 1974	17
DES 237,453	November 4, 1975	14
4,108,028	August 22, 1978	17
30,342	July 22, 1980	17
4,271,729	June 9, 1981	17
4,315,444	February 16, 1982	17

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, AT, GL, MI
Navy - AS, OS, SH
Air Force - 11, 17, 80, 90, 99
DLA - ES

User activities:
Army - AV, ME
Navy - MC, YD
Air Force - 13, 14, 19

Preparing activity:
Air Force - 85

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DLA - ES

(Project 6010-0025)

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